MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 9: Examples of simple complex integrals

Nick Strigul
Washington State University Vancouver
nick.strigul@wsu.edu

February 10, 2023

Lecture outline

1. Examples

Example 1

Example

Evaluate

$$\int_C \cos 2z dz$$

if $C : z(t) = i - 5e^{3it}$ for $0 \leq t \leq \pi/2$.

$$\int_C \cos 2z dz = \frac{1}{2}[\sin 2(z(\frac{\pi}{2})) - \sin 2(z(0))] =$$

$$= \frac{1}{2}[\sin 2(i - 5e^{3i\frac{\pi}{2}}) - \sin 2(i - 5)]$$

Example 2

Example

Show that

$$\left| \int_C e^{Re\, z} dz \right| \leq 4\pi e^2$$

if C is the circle of radius 2 about the origin oriented counterclockwise.

Lenght of C is 2π
$|e^{Re\, z}|$ is bounded by $e^2 = M$

$$\left| \int_C e^{Re\, z} dz \right| \leq LM = 4\pi e^2$$

Example 3. Part 1.

Example

Evaluate

$$\int_C \bar{z}\,dz$$

from $z = 0$ to $z = 4 + 2i$ along the curve $C : z = t^2 + i\,t$.

First we find $z' = 2t + i$, then we compute the integral

$$\begin{aligned}
\int_C \bar{z}\,dz &= \int_0^2 (t^2 - it)(2t + i)dt \\
&= \int_0^2 (2t^3 + t) + i(-t^2)dt \\
&= [\frac{2}{4}t^4 + \frac{1}{2}t^2 + i(-\frac{1}{3}t^3)]_0^2 \\
&= \frac{1}{2}2^4 + \frac{1}{2}2^2 + i(-\frac{1}{3}2^3) \\
&= 10 - \frac{8}{3}i
\end{aligned}$$

Example 3. Part 2.

Example

Evaluate

$$\int_C \bar{z}dz$$

from $z = 0$ to $z = 4 + 2i$ along the line from $z = 0$ to $z = 2i$ and then the line from $z = 2i$ to $z = 4 + 2i$.

First we parameterize these lines:

1) $z_1(t) = 2it$ for $0 < t < 1$, and $z_1'(t) = 2i$

2) $z_2(t) = 4t + 2i$ for $0 < t < 1$, and $z_2'(t) = 4$

Then we compute the integral:

$$\int_C \bar{z}dz = \int_0^1 -2it(2i)dt + \int_0^1 (4t - 2i)4dt = 10 - 8i$$

Example 4. Part 1.

Example

Let $f(z) = z$ and C be any rectifiable, continuous curve from $z = \alpha$ to $z = \beta$. Using the definition of the complex integral show that

$$\int_C z dz = \frac{1}{2}\left(\beta^2 - \alpha^2\right).$$

(Hint: First take $\zeta_j = z_j$ and then take $\zeta_j = z_{j-1}$ in the definition.)

First we consider $\Delta : a = t_0, t_1, t_2, ..., t_n = b$ and let $z_j = z(t_j)$, where $\alpha = z(t_0)$ and $\beta = z(t_n)$.

Then by definition

$$\int_C z dz = \lim_{||\Delta|| \to 0} \sum_{j=1}^{n} z_j(z_j - z_{j-1}) = \lim_{||\Delta|| \to 0} \sum_{j=1}^{n} (z_j^2 - z_j z_{j-1})$$

$$\int_C z dz = \lim_{||\Delta|| \to 0} \sum_{j=1}^{n} z_{j-1}(z_j - z_{j-1}) = \lim_{||\Delta|| \to 0} \sum_{j=1}^{n} (z_{j-1} z_j - z_{j-1}^2)$$

Example 4. Part 2.

Example

Adding these integrals together we have

$$
\begin{aligned}
2\int_C z dz &= \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_{j-1}z_j - z_{j-1}^2) + \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_j^2 - z_j z_{j-1}) \\
&= \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_{j-1}z_j - z_{j-1}^2 + z_j^2 - z_j z_{j-1}) \\
&= \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_j^2 - z_{j-1}^2) \\
&= \lim_{||\Delta||\to 0}[z_1^2 - z_0^2 + z_2^2 - z_1^2 + z_3^2 - z_2^2 + \cdots + z_n^2] \\
&= \lim_{||\Delta||\to 0}[-z_0^2 + z_n^2] = \beta^2 - \alpha^2
\end{aligned}
$$

$$
\int_C z dz = \frac{1}{2}\left(\beta^2 - \alpha^2\right)
$$

Example 4. Part 2.

Example

Adding these integrals together we have

$$
\begin{aligned}
2\int_C zdz &= \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_{j-1}z_j - z_{j-1}^2) + \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_j^2 - z_j z_{j-1}) \\
&= \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_{j-1}z_j - z_{j-1}^2 + z_j^2 - z_j z_{j-1}) \\
&= \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_j^2 - z_{j-1}^2) \\
&= \lim_{||\Delta||\to 0}[z_1^2 - z_0^2 + z_2^2 - z_1^2 + z_3^2 - z_2^2 + \cdots + z_n^2] \\
&= \lim_{||\Delta||\to 0}[-z_0^2 + z_n^2] = \beta^2 - \alpha^2
\end{aligned}
$$

$$
\int_C zdz = \frac{1}{2}\left(\beta^2 - \alpha^2\right)
$$

Example 5. Part 2.

Example

Adding these integrals together we have

$$
\begin{aligned}
2\int_C z dz &= \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_{j-1}z_j - z_{j-1}^2) + \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_j^2 - z_j z_{j-1}) \\
&= \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_{j-1}z_j - z_{j-1}^2 + z_j^2 - z_j z_{j-1}) \\
&= \lim_{||\Delta||\to 0}\sum_{j=1}^{n}(z_j^2 - z_{j-1}^2) \\
&= \lim_{||\Delta||\to 0}[z_1^2 - z_0^2 + z_2^2 - z_1^2 + z_3^2 - z_2^2 + \cdots + z_n^2] \\
&= \lim_{||\Delta||\to 0}[-z_0^2 + z_n^2] = \beta^2 - \alpha^2
\end{aligned}
$$

$$
\int_C z dz = \frac{1}{2}\left(\beta^2 - \alpha^2\right)
$$

Example 6. Part 1.

Example

Evaluate $\oint_C \frac{2z+1}{z^3 - iz^2 + 6z} dz$ if C is the circle of radius $\frac{1}{3}$ about $3i$.

$$\oint_C \frac{2z+1}{z^3 - iz^2 + 6z} dz = \oint_C \frac{2z+1}{z(z^2 - iz + 6)} dz = \oint_C \frac{2z+1}{z(z-3i)(z+2i)} dz$$

$$\frac{2z+1}{z(z-3i)(z+2i)} = \frac{A}{z} + \frac{B}{z-3i} + \frac{C}{z+2i}$$

When we solve for A, B, and C, we get:

$A = \frac{1}{6}$

$B = \frac{-6}{15} i - \frac{1}{15}$

$C = \frac{2}{5} i - \frac{1}{10}$

Therefore

$$\oint_C \frac{2z+1}{z(z-3i)(z+2i)} dz = \oint_C \frac{1}{6z} dz + \oint_C \frac{\left(\frac{-6}{15} i - \frac{1}{15}\right)}{z-3i} dz + \oint_C \frac{\frac{2}{5} i - \frac{1}{10}}{z+2i} dz$$

Example 6. Part 2.

Example

Since $z = 0$ and $z = -2i$ lie outside C, then

$$\oint_C \frac{1}{6z} dz = 0$$

and

$$\oint_C \frac{\frac{2}{5}i - \frac{1}{10}}{z + 2i} dz = 0$$

Therefore, we have to solve only the second integral:

$$C : z(t) = 3i + \tfrac{1}{3}e^{it} \qquad 0 \leq t \leq 2\pi$$

$$z'(t) = \tfrac{1}{3}ie^{it}$$

So we get,

$$\int_0^{2\pi} \frac{(\frac{-6}{15}i - \frac{1}{15})}{\frac{1}{3}e^{it}} \frac{1}{3} ie^{it} dt = [i(\frac{-6}{15}i - \frac{1}{15})t]_0^{2\pi} = 2\pi i(-\frac{6}{15}i - \frac{1}{15})$$

Thank you